2022 Midterm – Independent Corrected Solutions

Complete original questions 1–6, 100 points

Method. Derivations follow the printed topology and values. The graded handwriting and supplement are comparison sources; accepted marks do not validate a mistaken physical claim. All diagrams are editable TikZ, and the checks use graph/truth or stipulated RC models without process simulation.

Question 1: concepts (35 points)

(a) Mask transfer. Prepare the wafer, coat photoresist and soft bake; align the mask and expose; develop the resist; etch the exposed underlying material; strip the remaining resist. Exposure changes the resist and etching transfers its developed pattern into the substrate/layer. The handwritten outline is expanded to include development. [SFU fabrication lecture]

(b) SOI. An active silicon layer sits on buried oxide (BOX), above a silicon handle wafer. The insulating BOX isolates it from bulk silicon, reducing source/drain-to-bulk junction area and leakage. Fully depleted SOI additionally improves channel electrostatic control and can reduce off-state leakage. Source/body and drain/body PN junctions can remain in the film; the student's claim of no PN junctions is too strong. Gate and channel leakage are not automatically eliminated. [ST FD-SOI]

(c) Oxidation terminology. Two actual thermal oxidation methods are dry oxidation and wet oxidation:

$$\mathrm{Si} + \mathrm{O_2} \rightarrow \mathrm{SiO_2}, \qquad \mathrm{Si} + 2\mathrm{H_2O} \rightarrow \mathrm{SiO_2} + 2\mathrm{H_2}.$$

The supplied answers alternatively classify oxide formation as thermal oxidation and CVD oxide deposition. Thermal oxidation consumes substrate silicon; CVD deposits an oxide film. CVD is not direct oxidation of the silicon substrate. The original wording does not uniquely choose between thermal variants and oxide-formation processes, so both classifications are stated explicitly. [UTEP oxidation], [BYU growth/deposition]

(d) Static CMOS features. A valid steady-state input gives full rail-to-rail output restoration and ideally zero DC supply current, because one complementary network is OFF. Practical static power is leakage; both networks can conduct during switching. A single complementary complex gate is inverting, while multistage static CMOS circuits can be noninverting.

(e) Static power. Reverse-biased source/drain-to-body/well PN junctions produce junction leakage; OFF MOS transistors produce subthreshold channel leakage. $P_{static} = V_{DD} I_{leak}$. Replace the handwritten latch-up/BJT answer with subthreshold leakage; latch-up is an abnormal failure condition rather than the intended normal leakage component. [Berkeley CMOS power]

(f) Pull-up/down polarity. PMOS drives a strong logic 1 to V_{DD} and NMOS a strong logic 0 to GND. With gate voltages restricted to the supply rails, an NMOS pull-up loses about V_{tn} and a PMOS pull-down stops near $|V_{tp}|$. The normal complementary assignment gives full output swing.

(g) High-k. $C_{ox} = \epsilon_0 k / t_{phys}$. A larger dielectric constant allows a physically thicker insulator at the same gate capacitance/equivalent oxide thickness, reducing direct tunneling while maintaining gate control. Distinguish physical from electrical thickness. [Intel high-k]

Question 1 (continued)

(h) TSV processes. Via-first fabricates TSVs before CMOS FEOL device fabrication; via-last does so after FEOL and BEOL fabrication. Via-middle, a separate category, occurs after FEOL and before BEOL. Thus the student's "before BEOL" definition is insufficiently precise for via-first. [Burkett et al., TSV tutorial]

(i) Body effect. In a series nMOS stack, the device nearest the output may have the highest internal source voltage while its body stays at GND. Its source/body reverse bias increases its threshold:

$$V_{tn} = V_{tn0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right).$$

Larger threshold reduces overdrive and drive current. An analogous reverse source/well bias increases the pMOS threshold magnitude. Connection of a lone transistor drain to the output does not by itself cause severe body effect; this explanation concerns displaced source potentials in a stack. Correct the student's trend to $V_{SB} \uparrow \Rightarrow V_{tn} \uparrow$, with the body fixed at GND. [McMaster MOSFET lecture]

(j) Antenna effect. A floating conductor attached to a thin gate oxide can collect charge during plasma processing and damage the dielectric. Two fixes: (1) add an antenna protection diode/diffusion discharge path; (2) add a jumper near the gate, moving the long conductor to a later-formed higher metal layer so that a smaller area is attached during vulnerable fabrication steps. Arbitrarily splitting layers without the appropriate process-stage isolation is insufficient. [SkyWater rules], [Wang and Zhou jumper insertion]

(k) Bonding classifications. Orientation styles are face-to-face, face-to-back and back-to-back; face denotes the active-device/interconnect side and back the substrate side. A separate manufacturing-granularity classification is wafer-to-wafer, die-to-wafer and die-to-die. The original does not specify the classification axis. The graded sheet lists orientations and the supplement lists granularities; neither grading nor one list disproves the other. [Vaisband 3D integration], [imec assembly granularity]

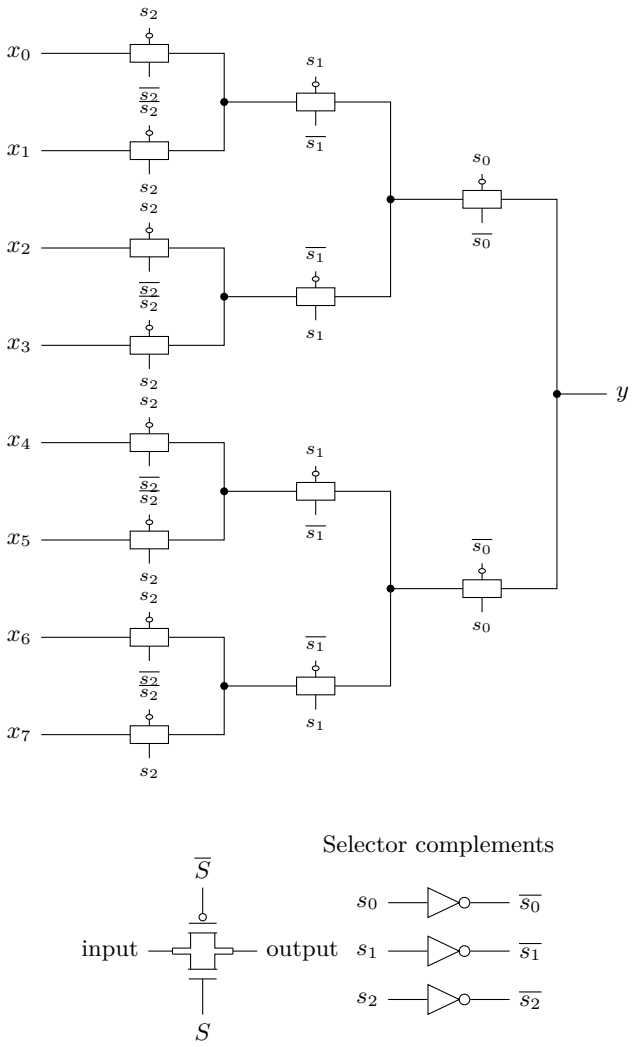
(l) Output capacitance. $C_L = C_{gate,load} + C_{diffusion} + C_{wire}$: receiver input gate capacitances, the driving gate's output-connected drain diffusion/junction capacitances, and routing capacitance to the substrate and neighboring wires. Both supplied lists are valid; receiver gate capacitance is meant here.

(m) FinFET types. Shorted-gate (SG): the two side gates are electrically tied. Independent-gate (IG): the side gates can be controlled separately. Correct the student's expansion "short gate" to "shorted-gate". [Chaudhuri and Jha, FinFET logic]

Primary references support the physical corrections; the exam source determines the exact question text and marks. The notes' Q1 lettering shifts after an added crosstalk answer, so they were matched by wording.

Question 2(a): 8-to-1 transmission-gate multiplexer

Choose the explicit index convention $k = 4s_0 + 2s_1 + s_2$, so $y = x_k$. Build seven 2-to-1 cells in a three-level binary tree: four cells controlled by s_2, two by s_1, and one by s_0. Each cell uses two transmission gates. For its input 0, enable on $\overline{s}$; for input 1, enable on s. Every TG consists of a parallel nMOS/pMOS pair. A bubble marks its pMOS gate.



TG detail and the three selector-complement generators.

Each selected data path contains three ON TGs; exactly one of $x_0, \ldots, x_7$ connects to y for a settled selection code. There are 14 TGs (28 pass transistors), plus the three CMOS inverters shown (6 transistors) to generate selector complements: 34 total. Tie pass nMOS bodies to GND and pMOS bodies to V_{DD}. The graded decoder-style arrangement can also select one input, but does not justify a linearly growing unbuffered delay.

Question 2(b): unbuffered delay scaling

With N selection bits there are 2^N data inputs and N successive TG levels along a selected tree path. Treat the ON TGs as R_{TG} and each visited-node parasitic as C_{int}; let external load be C_L. The Elmore delay scale is

$$\tau \simeq R_{TG}C_{int}\frac{N(N+1)}{2} + NR_{TG}C_L.$$

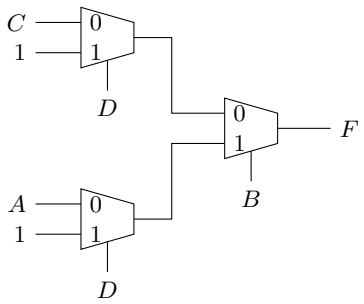
Upstream resistances must charge all downstream node capacitances, so internal RC delay grows quadratically with unbuffered depth: $\tau = O(N^2)$ when each node capacitance is comparable. For three levels, $\tau \simeq 6R_{TG}C_{int} + 3R_{TG}C_L$. A chain of restoring buffers is a different case, for which comparable stage delays can sum linearly. The graded sheet's $T \propto N$ claim omits the distributed capacitive loading; the supplement recognizes the square dependence. No absolute delay is determined because TG resistance and node/load capacitance values are not supplied.

Question 2(c): complete the three multiplexers

Read the printed 0/1 port labels; handwritten letters around Figure 1(b) are not constraints. Shannon expansion on B gives

$$F = \overline{B}(C + D) + B(A + D).$$

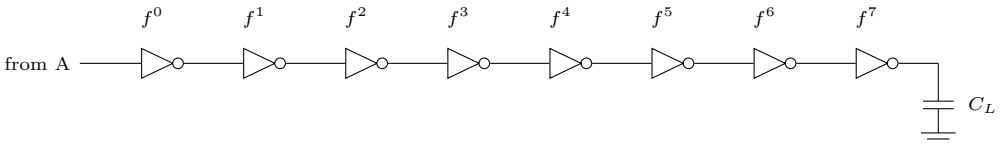
Use D to select between C and 1 for the upper mux, and between A and 1 for the lower mux. Select the upper/lower outputs with B:



For $B = 0$, the upper result is $C + D$; for $B = 1$, the lower result is $A + D$. Hence $F = AB + \overline{B}C + D$. This agrees with the graded completion after mapping the port order. The verifier checks all 16 input combinations.

Question 3(a): minimum-delay buffer with original equal-width devices

The load ratio is $H = C_L/C_g = 2.98\,\text{pF}/1\,\text{fF} = 2980$. Let N inverter stages have equal n/p width factors s_i relative to W_{min}, with $s_0 = 1$ so NAND A sees exactly C_g. The dashed buffer is noninverting, so N must be even. For fixed N, equal interstage ratios give $s_i = f^i$, where $f = H^{1/N}$; the last load is HC_g.



Exactly eight inverters; labels are width multipliers, not stage counts.

Each nMOS resistance is R/s_i and each equal-width pMOS is $2R/s_i$ because $\mu_n = 2\mu_p$. Over an even chain, either transition direction visits $N/2$ nMOS and $N/2$ pMOS pull paths. Thus

$$T_{unmodified} = \frac{3}{2} N H^{1/N} R C_g.$$

For arbitrary positive stage efforts f_i with $\prod_i f_i = H$, the two direction delays sum to $3RC_g \sum_i f_i$. Consequently their maximum is at least $\frac{3}{2} RC_g \sum_i f_i \geq \frac{3}{2} N H^{1/N} RC_g$ by AM–GM. Equal efforts attain this bound for an even chain, establishing the geometric taper for the worst-case objective despite unequal n/p mobility. The continuous minimum occurs near $N = \ln H \simeq 8$; comparing admissible even counts gives $\boxed{N = 8}$ and $f = 2980^{1/8} \simeq 2.71817 \simeq e$. Both n/p widths in stage $i = 0, \ldots, 7$ are $f^i W_{min}/L_{min}$. Exactly eight inverters are required: stage 0 through stage 7. The supplement's extra stage 8 makes nine inverters and reverses the required polarity. The graded sheet's eight-stage geometric taper is retained with explicit indexing and resistance accounting.

Question 3(b): fixed nMOS, balanced pMOS and changed loads

To make each inverter's rise/fall times equal, double its pMOS width while keeping its nMOS unchanged:

$$W_{n,i} = f^i W_{min}, \qquad W_{p,i} = 2f^i W_{min}.$$

With C_g defined as the total input capacitance of a minimum inverter whose two devices both have W_{min}, a single minimum transistor contributes $C_g/2$. The modified input gate capacitance is consequently

$$C_{in,i} = \frac{C_g}{2}(f^i + 2f^i) = \frac{3}{2} f^i C_g.$$

Both pull resistances are now R/f^i. The first seven stages drive modified next-stage gate loads; the eighth still drives the unchanged external C_L. Therefore

$$\boxed{T_{modified} = \left[\frac{3}{2}(N-1)+1\right] fRC_g = \frac{3N-1}{2} fRC_g.}$$

For $N = 8$, $T_{modified} = 11.5 fRC_g \simeq 31.25899 RC_g$. Using the paper's rounded $e = 2.7$ gives about $31.05 RC_g$. The last stage must not receive the 1.5 gate-load multiplier because C_L is fixed, external capacitance.

Input-load consequence. The first inverter now presents $1.5C_g$ to NAND A. Keeping its nMOS fixed, balancing the pMOS and keeping the original C_g input load cannot all hold simultaneously. Part(b) specifies the fixed nMOS modification; the changed upstream load is stated rather than silently claiming part(a)'s first-load condition remains satisfied. Upstream NAND delay is outside the source's marked buffer delay T.

The handwritten $T = NRC$ omits geometric stage effort and the altered gate loads. The verifier records all eight actual widths, gate capacitances and load terms; it also compares even stage counts for part(a). No new process-dependent timing parameters are assumed.

Question 4: Schmitt-type n-side feedback

Before output fall, $V_{out} = 1.2\,\text{V}$ turns feedback N2 on. The source topology and waveform are:

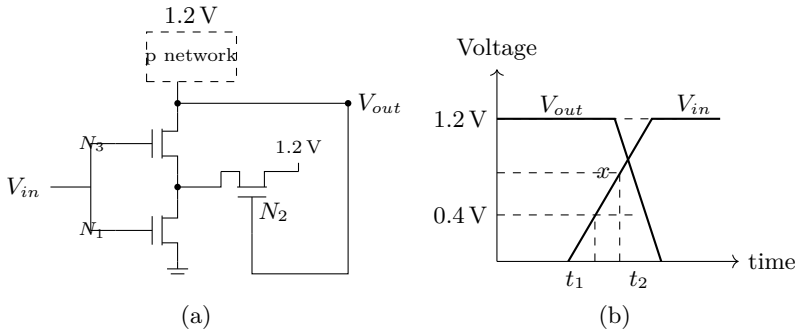


Figure 3: (a) A CMOS logic gate. (b) Waveforms of V_{in} and V_{out}.

With the stated ON resistances, N1 and N2 set their common internal node to

$$V_a = 1.2\frac{R}{R+2R} = 0.4\,\text{V}.$$

The upper series device N3 begins conducting when $V_{in} - V_a = V_{tn} = 0.4\,\text{V}$, giving

$$\boxed{x = V_{in}(t_2) = 0.8\,\text{V}.}$$

At t_1, $V_{in} = 0.4\,\text{V}$ and N1 reaches threshold. For the open interval $t_1 < t < t_2$:

Device	V_{GS}	V_{DS}	Region
N1	V_{in}	0.4 V	Saturation: $V_{DS} \geq V_{GS} - V_{tn}$
N2	0.8 V	0.8 V	Saturation
N3	$V_{in} - 0.4\,\text{V}$	0.8 V	Cutoff: $V_{GS} < V_{tn}$

At t_2, N3 is at threshold and N1 at the saturation/triode boundary; conduction then triggers output fall. These are the source's constant-ON-resistance estimates, not an exact MOS transient calculation. The supplied .8V and saturation/saturation/cutoff conclusions agree after fixing inconsistent intermediate voltage labels. N2 has $V_{GS} = V_{DS} = .8V$, not an overdrive of.8V; its overdrive is.4V.

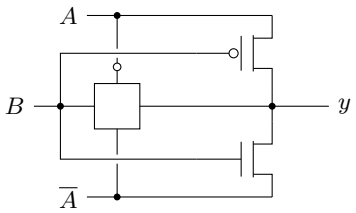
Question 5: the rail-fed/pass XOR

The left TG is enabled by $\overline{A}$: pMOS gate A, nMOS gate $\overline{A}$. The right pair has common gate B, pMOS source A, and nMOS source $\overline{A}$. Thus

$$A = 0 : \ y = B, \qquad A = 1 : \ y = \overline{B}, \qquad \boxed{y = A \oplus B}.$$

A	B	y	Valid settled drive path
0	0	0	TG connects $B = 0$; right pMOS also connects $A = 0$
0	1	1	TG passes $B = 1$ strongly; right nMOS connects $\overline{A} = 1$
1	0	1	TG off; right pMOS connects $A = 1$
1	1	0	TG off; right nMOS connects $\overline{A} = 0$

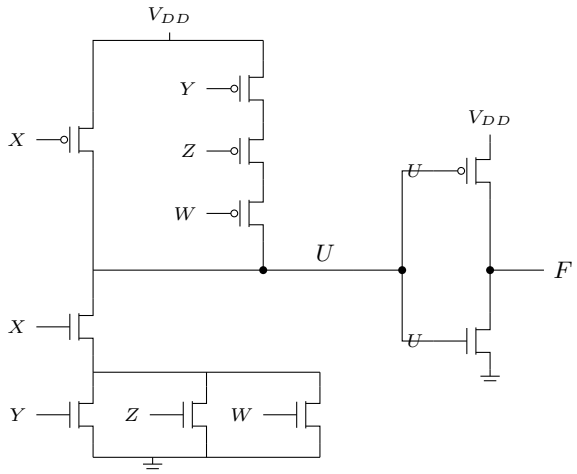
The complete TG avoids a threshold-loss problem for the two $A = 0$ cases; the active right device passes a strong 1 or 0 for $A = 1$. Under ideal settled input rails, all four outputs are correct.



Weakness. There is no separate restoring stage supplied directly by fixed V_{DD}/GND. Output drive comes through the pass network or through A/$\overline{A}$ input drivers acting as rails. Output resistance, fan-out capability and noise propagation depend on those input drivers; switched sources/body bias and finite input/complement arrival times can worsen delay or create transient glitches. The TG also provides no input isolation. The student's statement "no driving ability" is too strong: finite conductive paths exist, but it lacks independent full restoring drive. A restoring buffer pair could preserve XOR polarity while increasing load-driving capability; no extra devices are needed to identify the requested function and weaknesses.

Question 6(a): positive function $F = X(Y + Z) + XW$

Factor $F = X(Y + Z + W)$. Build $U = \overline{X(Y + Z + W)}$ with nMOS X in series with parallel Y, Z, W, and the dual pMOS network; add a static output inverter to obtain $F = \overline{U}$.



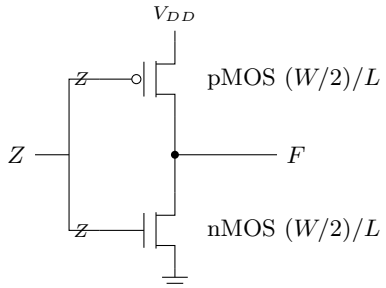
This requires 8 core devices plus 2 inverter devices, using only the original primary inputs. Tie all pMOS bodies to V_{DD} and all nMOS bodies to GND. The complemented-input student construction can be valid if explicit input inverters are included, as the supplement attempts; the factored original-input implementation avoids assuming free complements. The checker verifies all 16 assignments and PUN/PDN rail exclusivity.

Question 6(b): equivalent inverter for the specified transition

Here $F = \overline{X(Y+Z)}$, with every original transistor W/L. Fixing $X = 1, Y = 0$ leaves $F = \overline{Z}$ for the source sequence $101 \to 100 \to 101$, so $F : 0 \to 1 \to 0$.

Output transition	Conducting original path	Equivalent resistance
Rise ($101 \to 100$)	pMOS Y and Z in series	$2R_p$
Fall ($100 \to 101$)	nMOS X and Z in series	$2R_n$

Equivalent resistance doubles when width halves. An inverter driven by Z, with both its pMOS and nMOS width $W/2$ and unchanged length L, therefore represents these two active paths:



For symbolic output load C_o, $\boxed{t_r \simeq 2R_pC_o, \quad t_f \simeq 2R_nC_o}$. R_p, R_n refer to their original W/L devices. Q6 does not independently specify a mobility ratio or output capacitance, so these remain symbolic. If the $\mu_n = 2\mu_p$ convention is additionally applied, $R_p = 2R_n$ and $t_r = 2t_f$; this conditional ratio is not needed for the width-equivalence answer. This is an ON-resistance equivalent for the stated transition, not a claim of matching all internal capacitances or body biases. The handwritten half-width equivalent inverter agrees with the transition-specific derivation.

Reproducibility

Run `python verify.py`. Saved CSV/JSON checks cover eight mux selection codes, 16 Shannon inputs, RC ladder scaling, even buffer stage counts and each modified load, Schmitt-region sample points, four XOR routes, 16 CMOS truth states and the equivalent-inverter transition sequence. No simulated waveforms or device-model measurements are claimed.